

Q1. Consider a free particle constrained to move over the rectangular region $0 \leq x \leq a$, and $0 \leq y \leq b$. The eigenfunctions of this system are

$$f_{n_x n_y}(x, y) = \left(\frac{4}{ab}\right)^{1/2} \sin \frac{n_x \pi x}{a} \sin \frac{n_y \pi y}{b}$$

where, $n_x = 1, 2, 3, \dots$ and $n_y = 1, 2, 3, \dots$; the Hamiltonian operator for this system is

$$\hat{H} = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right)$$

Find the energy eigenvalues.

Ans

$$f(x, y) = f_{n_x} f_{n_y}(x, y) = \left(\frac{4}{ab}\right)^{1/2} \sin \frac{n_x \pi x}{a} \sin \frac{n_y \pi y}{b}$$

$$\hat{H} = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right)$$

$$= \underbrace{-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}}_{\hat{H}_x} - \underbrace{\frac{\hbar^2}{2m} \frac{\partial^2}{\partial y^2}}_{\hat{H}_y}$$

Find out

$$\hat{H}_x f(x, y) = ?$$

$$\hat{H}_y f(x, y) = ?$$

For $\hat{H}_x f(x, y)$

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \left[\left(\frac{4}{ab}\right)^{1/2} \sin \frac{n_x \pi x}{a} \sin \frac{n_y \pi y}{b} \right]$$

$$\Rightarrow +\frac{\hbar^2}{2m} \left(\frac{4}{ab}\right)^{1/2} \sin \frac{n_y \pi y}{b} \left[\left(\frac{n_x \pi}{a}\right)^2 \sin \frac{n_x \pi x}{a} \right]$$

Similarly
For $\hat{H}_y f(x, y)$

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial y^2} \left[\left(\frac{4}{ab}\right)^{1/2} \sin \frac{n_x \pi x}{a} \sin \frac{n_y \pi y}{b} \right]$$

$$\Rightarrow +\frac{\hbar^2}{2m} \left(\frac{4}{ab}\right)^{1/2} \sin \frac{n_x \pi x}{a} \left[\left(\frac{n_y \pi}{b}\right)^2 \sin \frac{n_y \pi y}{b} \right]$$

Now
for

$$\hat{H}_x f(x, y) + \hat{H}_y f(x, y)$$

$$\Rightarrow \frac{\hbar^2}{2m} \left(\frac{4}{ab}\right)^{1/2} \sin \frac{n_y \pi y}{b} \left[\left(\frac{n_x \pi}{a}\right)^2 \sin \frac{n_x \pi x}{a} \right] +$$

$$\frac{\hbar^2}{2m} \left(\frac{4}{ab}\right)^{1/2} \sin \frac{n_x \pi x}{b} \left[\left(\frac{n_y \pi}{a}\right)^2 \sin \frac{n_y \pi y}{a} \right]$$

$$\Rightarrow \frac{\hbar^2}{2m} \left[\left(\frac{n_x \pi}{a}\right)^2 + \left(\frac{n_y \pi}{a}\right)^2 \right] \times \underbrace{\left(\frac{4}{ab}\right)^{1/2} \sin \frac{n_x \pi x}{a} \sin \frac{n_y \pi y}{b}}_{f(x, y)}$$

$$\Rightarrow \frac{\hbar^2 \pi^2}{2m} \left[\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} \right] f(x, y)$$

$$\Rightarrow \underbrace{\frac{\hbar^2}{8m} \left[\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} \right]}_{\text{Energy eigenvalues}} f(x, y) \quad \Bigg| \quad A_{n_x, n_y}, \quad \hbar = \frac{h}{2\pi}$$

Q2. Specify which of the following operators are Hermitian: $-i\frac{d}{dx}$, $\frac{d^2}{dx^2}$, and $i\frac{d^2}{dx^2}$. Assume, that $0 \leq x \leq L$, and the functions on which these operators operate are appropriately well behaved:

$$(i) f(x) = \sin \frac{\pi x}{L}, \quad \text{where } f(0) = f(L) = 0$$

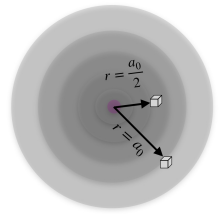
$$(ii) f(x) = e^{ix/L}, \quad \text{where } f(0) = f(L)$$

A2 Eigenvalues of any Hermitian operator are always real if they satisfy eigenvalue equation

Therefore,

<u>CASE</u>	<u>OPERATOR</u>	<u>Is it Hermitian?</u>
(i)	$i \frac{d}{dx}$	X
	$\frac{d^2}{dx^2}$	✓
	$i \frac{d^2}{dx^2}$	X
(ii)	$i \frac{d}{dx}$	✓
	$\frac{d^2}{dx^2}$	✓
	$i \frac{d^2}{dx^2}$	X

- The wavefunction of an electron in the lowest energy state of a He^+ ion is proportional to e^{-2r/a_0} , where r is the distance from the nucleus.
- Q3. Calculate the relative probabilities of finding the electron inside a small cubic volume located at a distance: (a) $r = \frac{a_0}{2}$, and (b) $r = a_0$.



A3

$$\psi \propto e^{-2r/a_0}$$

$$\text{Probability: } P = \int_0^x \psi^* \psi \, dx$$

$$\frac{P_{r=a_0/2}}{P_{r=a_0}} = \frac{\int_0^{a_0/2} e^{-\cancel{2}x\cancel{a_0}/\cancel{a_0}} e^{-\cancel{2}x\cancel{a_0}/\cancel{a_0}} \, dx}{\int_0^{a_0} e^{-\cancel{2}x\cancel{a_0}/\cancel{a_0}} e^{-\cancel{2}x\cancel{a_0}/\cancel{a_0}} \, dx}$$

$$= \frac{\int_0^{a_0/2} e^{-1} e^{-1} \, dx}{\int_0^{a_0} e^{-2} e^{-2} \, dx} = \frac{e^{-2}}{e^{-4}} \times \frac{\int_0^{a_0/2} dx}{\int_0^{a_0} dx}$$

$$\Rightarrow \frac{P_{r=a_0/2}}{P_{r=a_0}} = \frac{0.14}{0.02} \approx \frac{14\%}{2\%}$$

At $r = a_0$ probability of finding the electron reduces to 2% from 14% at $r = a_0/2$